

Package ‘dppca’

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Title Differentially Private Scree and Score Visualization

Version 0.1.0

Description Tools for performing principal component analysis (PCA) and constructing differentially private scree estimates and score-based visualizations. The package provides non-private and private PCA directions, DP scree estimators based on clipped means, Huber-type private means, and private winsorized means, as well as 2D score histograms with additive and sparse DP mechanisms, including group-wise visualizations.

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URL <https://github.com/yejinjo0220/dppca>, <https://yejinjo0220.github.io/dppca/>

BugReports <https://github.com/yejinjo0220/dppca/issues>

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adult	<i>Adult data example</i>
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Description

A sample dataset used for examples in the dppca package.

Usage

```
data(adult)
```

Format

A data frame or matrix.

Source

Prepared for examples in the dppca package.

clipped_control	<i>Control options for clipped scree estimation</i>
-----------------	---

Description

Creates a control list for the clipped-mean scree estimator used by `dp_scree()` and `dp_scree_plot()` when `method = "clipped"`.

Usage

```
clipped_control(C_clip)
```

Arguments

C_clip	Positive clipping threshold for squared principal component scores. This value has no default because it should be chosen according to the scale of the data.
--------	---

Details

The clipped method estimates each scree value by clipping the squared scores at `C_clip` and then applying a sensitivity-calibrated Gaussian mechanism (Dwork and Roth 2014). Larger values of `C_clip` reduce clipping bias but increase the sensitivity, and therefore the scale of the privacy noise.

Value

A list of control options for `method = "clipped"`.

References

Dwork C, Roth A (2014). "The Algorithmic Foundations of Differential Privacy." *Found. Trends Theor. Comput. Sci.*, **9**(3–4), 211–407. ISSN 1551-305X, doi:[10.1561/04000000042](https://doi.org/10.1561/04000000042).

See Also

[dp_scree\(\)](#) for using clipped scree estimation. [dp_scree_plot\(\)](#) for plotting private scree estimates.

Examples

```
clipped_control(C_clip = 3)
```

dppca_app

Launch the dppca Shiny app

Description

Opens an interactive Shiny app for exploring `dp_scree_plot()` and `dp_score_plot()` with the updated scree and score method options. If `X` is supplied, the app opens using that data. If `group` is supplied, the score plot uses the grouped version when the group labels are available. If `group = "color"`, the column named `color` is used as group labels and removed from the PCA feature matrix. In the app, the scree menu uses checkboxes and can overlay `method = "clipped"`, `method = "pmwm"`, `method = "huber"`, or a vector such as `method = c("clipped", "pmwm", "huber")`. The score menu can pass `method = "add"`, `method = "sparse"`, or `method = c("add", "sparse")` and bin counts through `bins = c(m_x, m_y)` to the score plotting functions. The app uses a taller, centered scree plot panel so that the scree curves are not visually flattened on wide screens.

Usage

```
dppca_app(X = NULL, group = NULL)
```

Arguments

<code>X</code>	Optional numeric matrix or data frame. If supplied, the app opens with this data selected as the default data source.
<code>group</code>	Optional group labels. This can be either a vector of length <code>nrow(X)</code> or a single column name in <code>X</code> .

Value

Invisibly launches a Shiny application.

Examples

```
## Not run:
dppca_app()
dppca_app(gau)
dppca_app(gau_g, group = "color")

## End(Not run)
```

dp_pc_dir

Estimate principal component directions

Description

This function returns the principal component directions used by the plotting and estimation functions in this package. By default, it computes the usual non-private directions from the sample covariance matrix. If `g_dppca = TRUE`, it computes private directions using the spherical-transformation version of g-DPPCA proposed by Kim and Jung (2025): it adds Gaussian noise to the spherical Kendall matrix and then extracts its leading eigenvectors.

Usage

```
dp_pc_dir(
  X,
  k,
  center = TRUE,
  standardize = FALSE,
  g_dppca = FALSE,
  eps_dir = NULL,
  delta_dir = NULL,
  cpp.option = FALSE
)
```

Arguments

<code>X</code>	A numeric matrix or data frame. Rows correspond to observations and columns correspond to variables.
<code>k</code>	Number of principal component directions to return. Must be an integer between 1 and the number of columns in <code>X</code> .
<code>center</code>	A logical value indicating whether to center the columns of <code>X</code> before computing principal component directions. The default is <code>TRUE</code> .
<code>standardize</code>	A logical value indicating whether to scale the columns of <code>X</code> by their sample standard deviations after optional centering. The default is <code>FALSE</code> .
<code>g_dppca</code>	Whether to compute private principal component directions using the spherical Kendall mechanism based on the g-DPPCA method. If <code>FALSE</code> , the usual non-private directions are computed from the sample covariance matrix. The default is <code>FALSE</code> .

eps_dir	Positive number defining the epsilon privacy parameter for private principal component directions. Required when g_dppca = TRUE.
delta_dir	Number in $(0, 1)$ defining the delta privacy parameter for private principal component directions. Required when g_dppca = TRUE.
cpp.option	A logical value reserved for a future C++ implementation of the spherical Kendall matrix. Currently only FALSE is supported.

Details

The non-private option computes leading eigenvectors of the sample covariance matrix of the pre-processed data. The private option is based on the spherical Kendall mechanism of Kim and Jung (2025): it first forms the spherical Kendall matrix from pairwise normalized differences, adds symmetric Gaussian noise, and then computes leading eigenvectors. The final eigenvector matrix is re-orthonormalized by QR decomposition.

Value

A numeric matrix with $\text{ncol}(X)$ rows and k columns. The columns are orthonormal principal component directions.

References

Kim M, Jung S (2025). “Robust and Differentially Private Principal Component Analysis.” *Statistical Analysis and Data Mining: An ASA Data Science Journal*, **18**(6), e70053. doi:10.1002/sam.70053, <https://onlinelibrary.wiley.com/doi/pdf/10.1002/sam.70053>, <https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.70053>.

Examples

```
data(gau, package = "dppca")

# Use a small subset to keep the example fast.
X <- head(gau, 50)

# Non-private principal component directions
V <- dp_pc_dir(X, k = 2)
head(V)

# Private principal component directions
set.seed(123)
V_private <- dp_pc_dir(
  X,
  k = 2,
  g_dppca = TRUE,
  eps_dir = 1,
  delta_dir = 1e-2
)
head(V_private)

# Generate a small low-rank dataset.
n <- 50
z1 <- rnorm(n)
z2 <- rnorm(n)
X <- cbind(
  x1 = z1 + 0.2 * rnorm(n),
```

```

x2 = 0.8 * z1 + 0.2 * rnorm(n),
x3 = z2 + 0.2 * rnorm(n),
x4 = 0.5 * z1 - 0.4 * z2 + 0.2 * rnorm(n)
)

# Non-private principal component directions
V <- dp_pc_dir(X, k = 2)
head(V)

# Private principal component directions
V_private <- dp_pc_dir(
  X,
  k = 2,
  g_dppca = TRUE,
  eps_dir = 1,
  delta_dir = 1e-2
)
head(V_private)

```

dp_score

Differentially private score histograms

Description

This function computes two-dimensional principal component scores and returns differentially private histogram estimates on the score space. It returns the score coordinates, the plotting frame, the non-private histogram, and the requested private histogram estimates.

Usage

```

dp_score(
  X,
  eps,
  delta,
  bins,
  method = c("add", "sparse"),
  center = TRUE,
  standardize = FALSE,
  g_dppca = FALSE,
  cpp.option = FALSE,
  axes = c(1, 2),
  fixed_frame = NULL
)

```

Arguments

X	A numeric matrix or data frame. Rows correspond to observations and columns correspond to variables.
eps	Positive number defining the total epsilon privacy parameter.
delta	Number in (0, 1) defining the total delta privacy parameter.

bins	Integer vector of length 2 defining the number of histogram bins along the first and second score axes, respectively.
method	Character vector specifying which private histogram methods to compute. Use "add" for the additive Gaussian histogram and "sparse" for the sparse thresholded histogram. The default is <code>c("add", "sparse")</code> .
center	A logical value indicating whether to center the columns of X before computing principal component directions. The default is TRUE.
standardize	A logical value indicating whether to scale the columns of X by their sample standard deviations after optional centering. The default is FALSE.
g_dppca	A logical value indicating whether to use private principal component directions. The default is FALSE. See <code>dp_pc_dir()</code> for details.
cpp.option	A logical value passed to <code>dp_pc_dir()</code> when <code>g_dppca = TRUE</code> . The default is FALSE.
axes	Integer vector of length 2 specifying the principal components used to construct the score coordinates. The default is <code>c(1, 2)</code> .
fixed_frame	Optional fixed plotting frame. If supplied, it can be a numeric vector <code>c(lower, upper)</code> used for both axes, or a list with numeric components <code>xlim</code> and <code>ylim</code> . If NULL, a private square frame is estimated from the score coordinates.

Details

Let v_a and v_b be the principal component directions selected by `axes = c(a, b)`. After preprocessing, the score point for observation i is $s_i = (x_i^\top v_a, x_i^\top v_b)$. A non-private score plot would display the points $s_1, \dots, s_n$ directly. This function instead summarizes their empirical distribution by a two-dimensional histogram and releases private versions of the histogram.

If `fixed_frame = NULL`, the plotting frame is constructed privately. The two score coordinates are stacked into one vector, private lower and upper quantiles are estimated using a smooth-sensitivity based quantile mechanism (Nissim et al. 2007), and the resulting interval is used as a common square frame for both axes. If `fixed_frame` is supplied, it is treated as public and no privacy budget is spent on frame construction.

The private histogram is computed on the rectangular grid defined by `fixed_frame` or by the private frame and the bin counts in `bins`. Under row-level adjacency, changing one observation can increase one bin count by one and decrease another by one, giving ℓ_1 sensitivity at most 2 and ℓ_2 sensitivity at most $\sqrt{2}$ for the count vector.

Two private histogram mechanisms are supported:

- "add" constructs an additive differentially private histogram by adding Gaussian noise to all bin counts, clipping negative noisy counts to zero, and normalizing the result. This additive-noise approach is commonly used for private histograms; see Wasserman and Zhou (2010).
- "sparse" constructs a sparse differentially private histogram for settings where many bins are empty. It perturbs only nonzero empirical bin proportions and keeps bins whose noisy values exceed a stability threshold, following the stability-based private histogram idea of Karwa and Vadhan (2017).

The privacy parameters are allocated across the privacy-consuming steps. If `g_dppca = FALSE` and `fixed_frame = NULL`, half of `eps` and `delta` is used for private frame construction and half for the private histogram. If `g_dppca = TRUE` and `fixed_frame = NULL`, the parameters are split equally among private direction estimation, private frame construction, and private histogram release. If `fixed_frame` is supplied, the frame step is skipped and the remaining steps split the privacy parameters equally.

Value

A list with components:

score An $n \times 2$ matrix of score coordinates.

frame A list with components xlim and ylim.

none Data frame for the non-private empirical histogram.

add Data frame for the additive Gaussian private histogram, or NULL if not requested.

sparse Data frame for the sparse private histogram, or NULL if not requested.

method Character vector of private histogram methods used.

References

Dwork C, Roth A (2014). “The Algorithmic Foundations of Differential Privacy.” *Found. Trends Theor. Comput. Sci.*, **9**(3–4), 211–407. ISSN 1551-305X, doi:10.1561/04000000042.

Nissim K, Raskhodnikova S, Smith A (2007). “Smooth Sensitivity and Sampling in Private Data Analysis.” In *STOC’07: Proceedings of the 39th Annual ACM Symposium on Theory of Computing*, 75–84. ISBN 9781595936318, doi:10.1145/1250790.1250803.

Wasserman L, Zhou S (2010). “A Statistical Framework for Differential Privacy.” *Journal of the American Statistical Association*, **105**(489), 375–389. doi:10.1198/jasa.2009.tm08651.

Karwa V, Vadhan SP (2017). “Finite Sample Differentially Private Confidence Intervals.” *CoRR*, abs/1711.03908. 1711.03908.

Kim M, Jung S (2025). “Robust and Differentially Private Principal Component Analysis.” *Statistical Analysis and Data Mining: An ASA Data Science Journal*, **18**(6), e70053. doi:10.1002/sam.70053, <https://onlinelibrary.wiley.com/doi/pdf/10.1002/sam.70053>, <https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.70053>.

See Also

`dp_score_plot()` for plotting the output of this function. `dp_score_group()` and `dp_score_plot_group()` for group-wise score histograms. `dp_pc_dir()` for private principal component direction estimation.

Examples

```
data(gau, package = "dppca")

# Use a small subset to keep the example fast.
X <- head(gau, 50)

set.seed(123)
score_out <- dp_score(
  X,
  eps = 1,
  delta = 1e-2,
  bins = c(3, 3),
  method = "add"
)
head(score_out$none)
head(score_out$add)

# Simulated low-rank data.
set.seed(123)
```

```

n <- 50
z1 <- rnorm(n)
z2 <- rnorm(n)
X_sim <- cbind(
  x1 = z1 + 0.2 * rnorm(n),
  x2 = 0.8 * z1 + 0.2 * rnorm(n),
  x3 = z2 + 0.2 * rnorm(n),
  x4 = 0.5 * z1 - 0.4 * z2 + 0.2 * rnorm(n)
)

set.seed(123)
dp_score(X_sim, eps = 1, delta = 1e-2, bins = c(3, 3))

```

dp_score_group	<i>Group-wise differentially private score histograms</i>
----------------	---

Description

This function computes two-dimensional principal component scores and releases group-wise differentially private histograms on a common score frame and grid. It is useful when observations have group labels and the low-dimensional score distribution should be compared across groups.

Usage

```

dp_score_group(
  X,
  group,
  eps,
  delta,
  bins,
  method = c("add", "sparse"),
  center = TRUE,
  standardize = FALSE,
  g_dppca = FALSE,
  cpp.option = FALSE,
  axes = c(1, 2),
  fixed_frame = NULL
)

```

Arguments

X	A numeric matrix or data frame. Rows correspond to observations and columns correspond to variables.
group	Group labels. This can be a vector of length <code>nrow(X)</code> or a single column name in X. If a column name is supplied, that column is used as the group label and removed from the feature matrix.
eps	Positive number defining the total epsilon privacy parameter.
delta	Number in $(0, 1)$ defining the total delta privacy parameter.
bins	Integer vector of length 2 defining the number of histogram bins along the first and second score axes, respectively.

method	Character vector specifying which private histogram methods to compute. Use "add" for the additive Gaussian histogram and "sparse" for the sparse thresholded histogram. The default is <code>c("add", "sparse")</code> .
center	A logical value indicating whether to center the columns of <code>X</code> before computing principal component directions. The default is <code>TRUE</code> .
standardize	A logical value indicating whether to scale the columns of <code>X</code> by their sample standard deviations after optional centering. The default is <code>FALSE</code> .
g_dppca	A logical value indicating whether to use private principal component directions. The default is <code>FALSE</code> . See <code>dp_pc_dir()</code> for details.
cpp.option	A logical value passed to <code>dp_pc_dir()</code> when <code>g_dppca = TRUE</code> . The default is <code>FALSE</code> .
axes	Integer vector of length 2 specifying the principal components used to construct the score coordinates. The default is <code>c(1, 2)</code> .
fixed_frame	Optional fixed plotting frame. If supplied, it can be a numeric vector <code>c(lower, upper)</code> used for both axes, or a list with numeric components <code>xlim</code> and <code>ylim</code> . If <code>NULL</code> , a private square frame is estimated from the score coordinates.

Details

The score directions, plotting frame, and histogram grid are shared across all groups. For each group g , the group-specific count in bin B_k is $c_k^{(g)} = \sum_i 1\{s_i \in B_k, g_i = g\}$. Private histograms are then computed separately for each group on the common grid. Because the groups form a partition of the rows, the group-wise histograms use the same histogram privacy parameters for each group by parallel composition.

Value

A list with components:

score An $n \times 2$ matrix of score coordinates.

frame A list with components `xlim` and `ylim`.

groups A named list of group-specific histogram outputs.

method Character vector of private histogram methods used.

See Also

`dp_score_plot_group()` for plotting group-wise score histograms. `dp_score()` for pooled score histograms.

Examples

```
data(gau_g, package = "dppca")

X <- head(gau_g, 60)

set.seed(123)
group_out <- dp_score_group(
  X,
  group = "color",
  eps = 1,
  delta = 1e-2,
  bins = c(3, 3),
```

```

    method = "add"
  )
  names(group_out$groups)

```

dp_score_plot

Plot differentially private score histograms

Description

This function computes and visualizes two-dimensional principal component score histograms, including the original scatter plot, the non-private empirical histogram, and one or more differentially private histogram estimates. It is a plotting wrapper around [dp_score\(\)](#) and returns both the computed score output and ggplot objects.

Usage

```

dp_score_plot(
  X,
  eps,
  delta,
  bins,
  method = c("add", "sparse"),
  center = TRUE,
  standardize = FALSE,
  g_dppca = FALSE,
  cpp.option = FALSE,
  axes = c(1, 2),
  fixed_frame = NULL
)

```

Arguments

<code>X</code>	A numeric matrix or data frame. Rows correspond to observations and columns correspond to variables.
<code>eps</code>	Positive number defining the total epsilon privacy parameter.
<code>delta</code>	Number in $(0, 1)$ defining the total delta privacy parameter.
<code>bins</code>	Integer vector of length 2 defining the number of histogram bins along the first and second score axes, respectively.
<code>method</code>	Character vector specifying which private histogram methods to compute. Use "add" for the additive Gaussian histogram and "sparse" for the sparse thresholded histogram. The default is <code>c("add", "sparse")</code> .
<code>center</code>	A logical value indicating whether to center the columns of <code>X</code> before computing principal component directions. The default is <code>TRUE</code> .
<code>standardize</code>	A logical value indicating whether to scale the columns of <code>X</code> by their sample standard deviations after optional centering. The default is <code>FALSE</code> .
<code>g_dppca</code>	A logical value indicating whether to use private principal component directions. The default is <code>FALSE</code> . See dp_pc_dir() for details.

cpp.option	A logical value passed to <code>dp_pc_dir()</code> when <code>g_dppca = TRUE</code> . The default is FALSE.
axes	Integer vector of length 2 specifying the principal components used to construct the score coordinates. The default is <code>c(1, 2)</code> .
fixed_frame	Optional fixed plotting frame. If supplied, it can be a numeric vector <code>c(lower, upper)</code> used for both axes, or a list with numeric components <code>xlim</code> and <code>ylim</code> . If NULL, a private square frame is estimated from the score coordinates.

Value

A list with components:

score The output of `dp_score()`.

plot A list containing the scatter plot, histogram panels, and the combined patchwork plot.

See Also

`dp_score()` for computing score histograms without plotting. `dp_score_plot_group()` for group-wise score histogram plots.

Examples

```
data(gau, package = "dppca")
X <- head(gau, 50)

set.seed(123)
p <- dp_score_plot(
  X,
  eps = 1,
  delta = 1e-2,
  bins = c(3, 3),
  method = "add"
)
p$plot$all
```

dp_score_plot_group	<i>Plot group-wise differentially private score histograms</i>
---------------------	--

Description

This function computes and visualizes group-wise differentially private score histograms. It is a plotting wrapper around `dp_score_group()` and returns both the computed group-wise score output and ggplot objects.

Usage

```
dp_score_plot_group(
  X,
  group,
  eps,
  delta,
```

```

    bins,
    center = TRUE,
    standardize = FALSE,
    g_dppca = FALSE,
    cpp.option = FALSE,
    axes = c(1, 2),
    method = c("add", "sparse"),
    fixed_frame = NULL
  )

```

Arguments

<code>X</code>	A numeric matrix or data frame. Rows correspond to observations and columns correspond to variables.
<code>group</code>	Group labels. This can be a vector of length <code>nrow(X)</code> or a single column name in <code>X</code> . If a column name is supplied, that column is used as the group label and removed from the feature matrix.
<code>eps</code>	Positive number defining the total epsilon privacy parameter.
<code>delta</code>	Number in $(0, 1)$ defining the total delta privacy parameter.
<code>bins</code>	Integer vector of length 2 defining the number of histogram bins along the first and second score axes, respectively.
<code>center</code>	A logical value indicating whether to center the columns of <code>X</code> before computing principal component directions. The default is <code>TRUE</code> .
<code>standardize</code>	A logical value indicating whether to scale the columns of <code>X</code> by their sample standard deviations after optional centering. The default is <code>FALSE</code> .
<code>g_dppca</code>	A logical value indicating whether to use private principal component directions. The default is <code>FALSE</code> . See dp_pc_dir() for details.
<code>cpp.option</code>	A logical value passed to dp_pc_dir() when <code>g_dppca = TRUE</code> . The default is <code>FALSE</code> .
<code>axes</code>	Integer vector of length 2 specifying the principal components used to construct the score coordinates. The default is <code>c(1, 2)</code> .
<code>method</code>	Character vector specifying which private histogram methods to compute. Use <code>"add"</code> for the additive Gaussian histogram and <code>"sparse"</code> for the sparse thresholded histogram. The default is <code>c("add", "sparse")</code> .
<code>fixed_frame</code>	Optional fixed plotting frame. If supplied, it can be a numeric vector <code>c(lower, upper)</code> used for both axes, or a list with numeric components <code>xlim</code> and <code>ylim</code> . If <code>NULL</code> , a private square frame is estimated from the score coordinates.

Value

A list with components:

score The output of [dp_score_group\(\)](#).

plot A list containing group-wise histogram plots.

group_colors Named vector of colors used for the groups.

See Also

[dp_score_group\(\)](#) for computing group-wise score histograms without plotting. [dp_score_plot\(\)](#) for pooled score histogram plots.

Examples

```
data(gau_g, package = "dppca")

X <- head(gau_g, 60)

set.seed(123)
p <- dp_score_plot_group(
  X,
  group = "color",
  eps = 1,
  delta = 1e-5,
  bins = c(8, 8),
  method = "add"
)
p$plot$all
```

dp_scree

Differentially private scree estimates

Description

This function computes scree estimates for principal component analysis, including both the usual non-private estimates and differentially private estimates. The private estimates are computed from principal component scores using one of the supported private mean estimators. See Details for the estimating equations and method-specific construction.

Usage

```
dp_scree(
  X,
  k,
  method = c("clipped", "pmwm", "huber"),
  control = NULL,
  eps,
  delta,
  center = TRUE,
  standardize = FALSE,
  g_dppca = FALSE,
  cpp.option = FALSE,
  mono = TRUE
)
```

Arguments

X	A numeric matrix or data frame. Rows correspond to observations and columns correspond to variables.
k	Positive integer defining the number of leading principal components to estimate. Must be an integer between 1 and the number of columns in X.
method	Scree estimation method. One of "clipped", "pmwm", or "huber".

control	Optional method-specific control list created by <code>clipped_control()</code> , <code>pmwm_control()</code> , or <code>huber_control()</code> .
eps	Positive number defining the total epsilon privacy parameter. If <code>g_dppca = TRUE</code> , it is split between private direction estimation and private scree estimation.
delta	Number in $(0, 1)$ defining the total delta privacy parameter. If <code>g_dppca = TRUE</code> , it is split between private direction estimation and private scree estimation.
center	A logical value indicating whether to center the columns of X before computing principal component directions. The default is <code>TRUE</code> .
standardize	A logical value indicating whether to scale the columns of X by their sample standard deviations after optional centering. The default is <code>FALSE</code> .
g_dppca	A logical value indicating whether to use private principal component directions for scree estimation. The default is <code>FALSE</code> . See <code>dp_pc_dir()</code> for details.
cpp.option	A logical value passed to <code>dp_pc_dir()</code> when <code>g_dppca = TRUE</code> . The default is <code>FALSE</code> .
mono	A logical value indicating whether to apply monotone post-processing to the private scree vector. The default is <code>TRUE</code> .

Details

Let X denote the preprocessed data matrix and let v_l be the l th principal component direction. The l th score vector is $z_l = Xv_l$. The corresponding sample scree value can be written as

$$\hat{\lambda}_l = \frac{1}{n-1} \sum_{i=1}^n z_{il}^2 = \frac{n}{n-1} \left(\frac{1}{n} \sum_{i=1}^n w_{il} \right), \quad w_{il} = z_{il}^2.$$

Therefore, each scree value is estimated by privately estimating the mean of $w_{1l}, \dots, w_{nl}$ and multiplying by $n/(n-1)$.

The supported methods differ in how this private mean is estimated:

- "clipped" clips the squared scores w_{il} at `C_clip` and then applies the Gaussian mechanism (Dwork and Roth 2014). This is the simplest option but depends directly on the clipping threshold.
- "pmwm" uses the private modified winsorized mean approach of Ramsay and Spicker (2025), adapted from the accompanying Python implementation into R. It privately estimates tail cut-offs, winsorizes the squared scores w_{il} , and releases a noisy winsorized mean.
- "huber" uses a Huber-type private robust mean estimator based on noisy gradient descent, following Yu et al. (2024).

The argument `g_dppca` controls how the principal component directions are obtained. If `g_dppca = FALSE`, the directions are computed non-privately and the full privacy parameters `eps` and `delta` are used for private scree estimation. If `g_dppca = TRUE`, the directions are computed privately using `dp_pc_dir()`. In that case, the privacy parameters are split equally: `eps / 2` and `delta / 2` are used for private direction estimation, and the remaining `eps / 2` and `delta / 2` are used for private scree estimation. If `mono = TRUE`, the final monotone adjustment is a post-processing step and does not change the privacy guarantee.

Value

A list with components:

- `method`: scree estimation method.

- scree_np: non-private scree estimates.
- pve_np: non-private proportions of variance explained.
- scree: differentially private scree estimates.
- pve: differentially private proportions of variance explained.

References

Dwork C, Roth A (2014). “The Algorithmic Foundations of Differential Privacy.” *Found. Trends Theor. Comput. Sci.*, **9**(3–4), 211–407. ISSN 1551-305X, doi:10.1561/04000000042.

Ramsay K, Spicker D (2025). “Improved subsample-and-aggregate via the private modified win-sorized mean.” Code available at <https://github.com/12ramsake/PMWM>, 2501.14095, <https://arxiv.org/abs/2501.14095>.

Yu M, Ren Z, Zhou W (2024). “Gaussian differentially private robust mean estimation and inference.” *Bernoulli*, **30**(4), 3059–3088.

Kim M, Jung S (2025). “Robust and Differentially Private Principal Component Analysis.” *Statistical Analysis and Data Mining: An ASA Data Science Journal*, **18**(6), e70053. doi:10.1002/sam.70053, <https://onlinelibrary.wiley.com/doi/pdf/10.1002/sam.70053>, <https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.70053>.

See Also

`dp_pc_dir()` for principal component direction estimation. `clipped_control()`, `pmwm_control()`, and `huber_control()` for method-specific tuning parameters.

Examples

```
set.seed(123)
n <- 50
z1 <- rnorm(n)
z2 <- rnorm(n)
X <- cbind(
  x1 = z1 + 0.2 * rnorm(n),
  x2 = 0.8 * z1 + 0.2 * rnorm(n),
  x3 = z2 + 0.2 * rnorm(n),
  x4 = 0.5 * z1 - 0.4 * z2 + 0.2 * rnorm(n)
)

dp_scree(
  X,
  k = 2,
  method = "clipped",
  control = clipped_control(C_clip = 3),
  eps = 1,
  delta = 1e-2
)

dp_scree(
  X,
  k = 2,
  method = "pmwm",
  control = pmwm_control(a = 0, b = 20, trim_const = 10, eta = 0.01),
  eps = 1,
  delta = 1e-2
)
```

```

)

dp_scee(
  X,
  k = 2,
  method = "huber",
  control = huber_control(k_min_m2 = -10, k_max_m2 = 10, m2_frac = 1 / 4),
  eps = 1,
  delta = 1e-2
)

```

dp_scee_plot

Plot differentially private scree estimates

Description

This function computes and visualizes scree curves for principal component analysis, including the usual non-private curve and one or more differentially private estimates. It is a plotting wrapper around [dp_scee\(\)](#) and returns a ggplot object.

Usage

```

dp_scee_plot(
  X,
  k,
  method = c("clipped", "pmwm", "huber"),
  control = NULL,
  eps,
  delta,
  center = TRUE,
  standardize = FALSE,
  g_dppca = FALSE,
  cpp.option = FALSE,
  mono = TRUE,
  type = c("pve", "scree")
)

```

Arguments

X	A numeric matrix or data frame. Rows correspond to observations and columns correspond to variables.
k	Positive integer defining the number of leading principal components to estimate. Must be an integer between 1 and the number of columns in X.
method	Scree estimation method or methods to plot. One or more of "clipped", "pmwm", or "huber".
control	Optional method-specific control list, or a named list of control lists when multiple methods are requested. Use clipped_control() , pmwm_control() , and huber_control() .

eps	Positive number defining the total epsilon privacy parameter. If <code>g_dppca = TRUE</code> , it is split between private direction estimation and private scree estimation.
delta	Number in $(0, 1)$ defining the total delta privacy parameter. If <code>g_dppca = TRUE</code> , it is split between private direction estimation and private scree estimation.
center	A logical value indicating whether to center the columns of <code>X</code> before computing principal component directions. The default is <code>TRUE</code> .
standardize	A logical value indicating whether to scale the columns of <code>X</code> by their sample standard deviations after optional centering. The default is <code>FALSE</code> .
g_dppca	A logical value indicating whether to use private principal component directions for scree estimation. The default is <code>FALSE</code> . See <code>dp_pc_dir()</code> for details.
cpp.option	A logical value passed to <code>dp_pc_dir()</code> when <code>g_dppca = TRUE</code> . The default is <code>FALSE</code> .
mono	A logical value indicating whether to apply monotone post-processing to the private scree vector. The default is <code>TRUE</code> .
type	Quantity to plot. Use <code>"pve"</code> to plot proportions of variance explained and <code>"scree"</code> to plot raw scree values. The default is <code>"pve"</code> .

Details

This function is a plotting wrapper around `dp_scree()`. For each requested method, it computes a private scree estimate and overlays it with the corresponding non-private curve. When `type = "pve"`, the plotted quantity is the proportion of variance explained (PVE); when `type = "scree"`, the raw scree values are shown.

To plot multiple methods, pass a character vector to `method`. If a method requires tuning parameters, pass `control` as a named list, for example `control = list(clipped = clipped_control(), pmwm = pmwm_control(), huber = huber_control())`.

For the estimating equations, privacy-budget allocation, and method-specific construction, see `dp_scree()`.

Value

Invisibly returns a list with components:

- `nonprivate`: non-private scree and PVE values.
- `results`: method-specific `dp_scree()` outputs used in the plot.

References

Dwork C, Roth A (2014). "The Algorithmic Foundations of Differential Privacy." *Found. Trends Theor. Comput. Sci.*, **9**(3–4), 211–407. ISSN 1551-305X, doi:10.1561/04000000042.

Ramsay K, Spicker D (2025). "Improved subsample-and-aggregate via the private modified win-sorized mean." Code available at <https://github.com/12ramsake/PMWM>, 2501.14095, <https://arxiv.org/abs/2501.14095>.

Yu M, Ren Z, Zhou W (2024). "Gaussian differentially private robust mean estimation and inference." *Bernoulli*, **30**(4), 3059–3088.

Kim M, Jung S (2025). "Robust and Differentially Private Principal Component Analysis." *Statistical Analysis and Data Mining: An ASA Data Science Journal*, **18**(6), e70053. doi:10.1002/sam.70053, <https://onlinelibrary.wiley.com/doi/pdf/10.1002/sam.70053>, <https://onlinelibrary.wiley.com/doi/abs/10.1002/sam.70053>.

See Also

[dp_pc_dir\(\)](#) for principal component direction estimation. [dp_scree\(\)](#) for computing non-private and differentially private scree estimates. [clipped_control\(\)](#), [pmwm_control\(\)](#), and [huber_control\(\)](#) for method-specific tuning parameters.

Examples

```
data(gau, package = "dppca")

X <- head(gau, 50)

dp_scree_plot(
  X,
  k = 2,
  method = "clipped",
  control = clipped_control(C_clip = 3),
  eps = 1,
  delta = 1e-2
)

dp_scree_plot(
  X,
  k = 2,
  method = c("clipped", "pmwm", "huber"),
  control = list(
    clipped = clipped_control(C_clip = 3),
    pmwm = pmwm_control(a = 0, b = 20, trim_const = 10, eta = 0.01),
    huber = huber_control(k_min_m2 = -10, k_max_m2 = 10, m2_frac = 1 / 4)
  ),
  eps = 1,
  delta = 1e-2
)
```

eur_map

Europe map PCA data

Description

A dataset used for PCA score plot examples.

Usage

```
data(eur_map)
```

Format

A data frame or matrix.

Source

Prepared for examples in the dppca package.

eur_map_g	<i>Group labels for Europe map data</i>
-----------	---

Description

Group labels corresponding to observations in eur_map.

Usage

```
data(eur_map_g)
```

Format

A vector or data frame containing group labels.

Source

Prepared for examples in the dppca package.

gau	<i>Gaussian example data</i>
-----	------------------------------

Description

A simulated Gaussian dataset used for examples.

Usage

```
data(gau)
```

Format

A data frame or matrix.

Source

Simulated for examples in the dppca package.

gau_g

*Group labels for Gaussian example data***Description**

Group labels corresponding to observations in gau.

Usage

```
data(gau_g)
```

Format

A vector or data frame containing group labels.

Source

Simulated for examples in the dppca package.

huber_control

*Control options for Huber scree estimation***Description**

Creates a control list for the Huber-type private scree estimator used by [dp_scree\(\)](#) and [dp_scree_plot\(\)](#) when method = "huber".

Usage

```
huber_control(
  k_min_m2,
  k_max_m2,
  m2_frac,
  mu0 = 0,
  eta0 = 1,
  T = NULL,
  M = NULL
)
```

Arguments

k_min_m2, k_max_m2

Integers defining the lower and upper dyadic bin indices used in the private second-moment scale step. The histogram searches over scale levels 2^k for $k_{\min} \leq k \leq k_{\max}$. These values have no defaults because they should be chosen according to the scale of the data.

m2_frac

Number in $(0, 1)$ defining the fraction of the Huber scree privacy parameters allocated to the private second-moment scale step. This value has no default.

mu0

Numeric initial value for Huber noisy gradient descent. The default is 0.

eta0	Positive number defining the fixed step size for Huber noisy gradient descent. The default is 1.
T	Optional positive integer defining the number of noisy gradient descent iterations. If NULL, the implementation uses $\lceil \log n \rceil$, where n is the number of observations.
M	Optional positive integer defining the number of blocks used in the private second-moment scale step. If NULL, the implementation uses $\lfloor \sqrt{n}/2 \rfloor$, where n is the number of observations.

Details

The Huber method estimates the mean of squared principal component scores by noisy gradient descent on the Huber loss. It follows the Huber-type private robust mean approach of Yu et al. (2024).

The method first privately estimates a scale proxy for the squared scores, denoted by m_2 . This scale proxy is then used to choose the Huber robustification level for noisy gradient descent. The parameters `k_min_m2`, `k_max_m2`, and `m2_frac` control this private scale-proxy step, while `mu0`, `eta0`, `T`, and `M` control the subsequent noisy gradient descent routine.

The dyadic indices `k_min_m2` and `k_max_m2` define the search range for the private histogram used to estimate m_2 . The histogram searches over candidate scale levels 2^k satisfying $k_{\min} \leq k \leq k_{\max}$. Because this range depends on the scale of the squared scores, these arguments are intentionally not given defaults.

The argument `m2_frac` determines how the Huber scree privacy parameters are split between the private scale-proxy step and the noisy gradient descent step. If $(\epsilon_{\text{scree}}, \delta_{\text{scree}})$ denotes the privacy parameters available for Huber scree estimation, then $m2_frac \cdot (\epsilon_{\text{scree}}, \delta_{\text{scree}})$ is used to privately estimate m_2 , while $(1 - m2_frac) \cdot (\epsilon_{\text{scree}}, \delta_{\text{scree}})$ is used for Huber noisy gradient descent.

The remaining parameters have default values. The default `mu0` = 0 is the initial value for noisy gradient descent, and the default `eta0` = 1 is the fixed step size. If `T` = NULL, the number of noisy gradient descent iterations is chosen as $\lceil \log n \rceil$. If `M` = NULL, the number of blocks used in the private estimator of m_2 is chosen as $\lfloor \sqrt{n}/2 \rfloor$.

Value

A list of control options for `method = "huber"`.

References

Yu M, Ren Z, Zhou W (2024). “Gaussian differentially private robust mean estimation and inference.” *Bernoulli*, **30**(4), 3059–3088.

See Also

[dp_scree\(\)](#) for computing differentially private scree estimates using these control options. [dp_scree_plot\(\)](#) for plotting scree estimates.

Examples

```
huber_control(k_min_m2 = -10, k_max_m2 = 10, m2_frac = 1 / 4)
huber_control(
  k_min_m2 = -10,
  k_max_m2 = 10,
  m2_frac = 1 / 4,
```

```

    T = 50,
    M = 20
)

```

pmwm_control

Control options for private modified winsorized scree estimation

Description

Creates a control list for the private modified winsorized mean scree estimator used by `dp_scree()` and `dp_scree_plot()` when `method = "pmwm"`.

Usage

```
pmwm_control(a, b, trim_const, eta, beta = 1.001, split_mode = TRUE)
```

Arguments

<code>a, b</code>	Finite lower and upper search bounds supplied to the private quantile routine. The private lower and upper clipping cutoffs are searched within this range. These values have no defaults because they should be chosen on the scale of squared principal component scores.
<code>trim_const</code>	Positive number controlling the baseline clipping level in the practical clipping proportion. This value has no default.
<code>eta</code>	Nonnegative number controlling the expected contamination level in the practical clipping proportion. This value has no default.
<code>beta</code>	Positive number greater than 1 defining the log-binning base used by the private quantile routine. The default is 1.001.
<code>split_mode</code>	A logical value indicating whether to split the sample into quantile-estimation and mean-estimation subsets. The default is TRUE.

Details

The PMWM method privately estimates lower and upper tail cutoffs, winsorizes the squared scores to those cutoffs, and then releases a noisy winsorized mean. It is based on the private modified winsorized mean of Ramsay and Spicker (2025).

The implementation used here is an R adaptation of the publicly available Python implementation accompanying Ramsay and Spicker (2025). The adaptation is used for scree estimation by applying the PMWM estimator to squared principal component scores.

The PMWM scree estimator uses additional control parameters for private quantile estimation and winsorization. The parameter `beta` determines the spacing of the geometric search grid used by the private quantile estimator and must satisfy $\beta > 1$. Smaller values of `beta` give a finer grid but may increase computation.

The bounds `a` and `b` define the lower and upper search range supplied to the private quantile routine. The private lower and upper winsorization cutoffs are searched within this range. These bounds should be chosen on the scale of the squared principal component scores.

The parameters `trim_const` and `eta` determine the practical clipping proportion used by the modified winsorized mean. If n_q denotes the number of observations used for private quantile estimation, the clipping proportion is

$$p = \min \left\{ \max \left(\frac{\text{trim_const}}{n_q}, \eta \right), 0.49 \right\}.$$

Here, `trim_const / n_q` controls the baseline clipping level, while `eta` gives a lower bound reflecting the expected contamination level.

If `split_mode = TRUE`, the sample is split into two parts: one part is used for private quantile estimation and the other part is used for the winsorized mean step. If `split_mode = FALSE`, all observations are used in both steps.

The parameters `a`, `b`, `trim_const`, and `eta` are intentionally not given defaults. They are data- and robustness-dependent choices and should be set deliberately by the user.

Value

A list of control options for `method = "pmwm"`.

References

Ramsay K, Spicker D (2025). "Improved subsample-and-aggregate via the private modified winsorized mean." Code available at <https://github.com/12ramsake/PMWM>, 2501.14095, <https://arxiv.org/abs/2501.14095>.

See Also

`dp_scree()` for computing differentially private scree estimates using these control options. `dp_scree_plot()` for plotting scree estimates.

Examples

```
pmwm_control(a = 0, b = 20, trim_const = 10, eta = 0.01)
pmwm_control(
  a = 0,
  b = 20,
  trim_const = 10,
  eta = 0.01,
  beta = 1.001,
  split_mode = FALSE
)
```

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